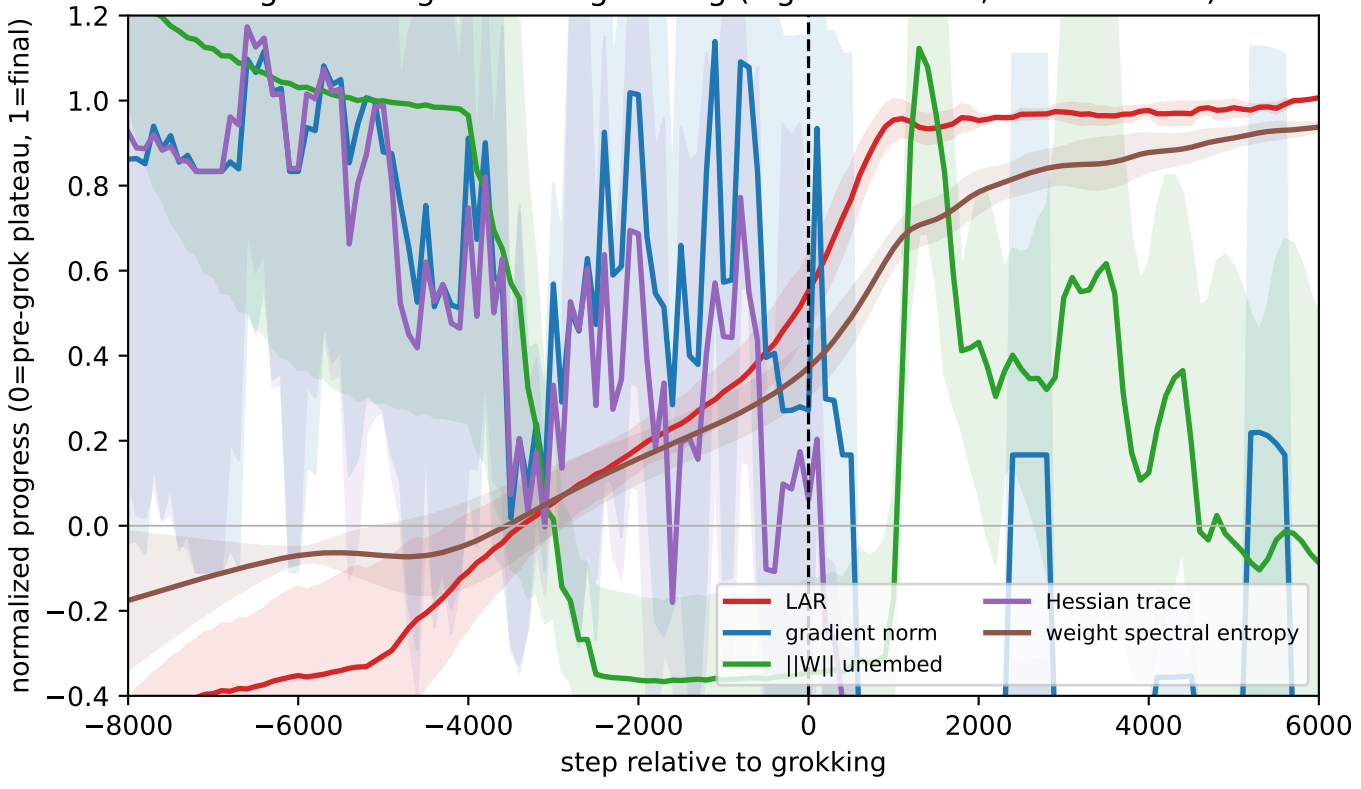
Signal change around grokking (sign-corrected, init excluded)
normalized progress (0=pre-grok plateau, 1=final)
step relative to grokking
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−8000
−6000
−4000
−2000
0
2000
4000
6000
LAR
gradient norm
||W|| unembed
Hessian trace
weight spectral entropy